Journal of Innovation & Knowledge 8 (2023) 100407

Journal of Innovation & Knowledge

https://www.journals.elsevier.com/journal-of-innovation-and-knowledge

# Global labour markets and workplaces in the age of intelligent machines

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ARTICLE INFO

*Article History:*
Received 21 June 2022
Accepted 29 June 2023
Available online 10 August 2023

*Keywords:*
Manufacturing innovation
Industrial innovation
Inclusive innovation
Intelligent machines
Human resources
Labour productivity
Future workplaces
Technology

ABSTRACT

This paper examines how labour productivity will increase in future workplaces that employ human resources and machine technologies. This paper expands the concept of intelligent machines as systems of automating labour based on robots and artificial intelligence. Furthermore, it aims to develop recommendations for improving productivity by managing competition and marketing in current global labour markets. We model the dependence of output per worker on robots' international distribution, the ease of hiring foreign labour, and the dependence on pay and productivity. As a result of this research, labour productivity is defined by human resource availability and not through automation. The study develops comprehensive guidance for economic policy in the global labour market for the medium-term period. Management implications include representing a vision for the future of workplaces drawn on the productivity benefits of different workplace models that stimulate growth in labour efficiency. The recommended concept validates how innovation and digitalisation reshape the workplace.

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## Introduction

The labour market is a complex social and economic relations system constantly changing. It thus needs to be studied and managed to consider the specifics of the current context in which this system operates and determine the future of work (Moffett et al., 2021). Knowledge-intensive and high-tech manufacturing is a priority for forming global competitive advantages; therefore, it is a vector of labour market growth (Al-Omoush et al., 2020; Puranam, 2021). A transformation of global labour markets and workplaces takes place and reshapes workplaces since more functions previously performed by human resources are passed on to intelligent machines (Schäfer et al., 2023).

Intelligent machines are means of automatization that rely on the leading technologies of Industry 4.0 – robots and artificial intelligence. Unlike an established concept of digital technologies undoubtedly linked to Industry 4.0, intelligent machines are treated as disruptive technologies that systematically change the character of human labour and reshape workplaces. Digital technologies complement human labour, and intelligent machines replace and oust (even wholly) human resources from entrepreneurial processes and systems.

Second, globalisation has vastly increased the accessibility of national labour markets. Global labour markets are characterised by the flow of international labour migration and the parallel involvement of foreign labour (Corella, 2020; Hober et al., 2021). Third, human resources are represented in the labour market and machine technologies (Schumacher, 2021). In the pre-digital period, it was common to assess the labour market from the points of view of quantitative sufficiency (lack of deficit) and the quality (level of qualification) of human resources (Ballestar et al., 2021; Camargo et al., 2021). However, human resources formed a fundamentally new conceptual approach to assessing the labour market from the productivity perspective, provided through technological support and automation tools (Boza & Ilyés, 2020). In this context, labour efficiency is defined not so much by human resources as by automation tools.

Despite fragmented research and general theoretical assumptions about both sources of productivity growth – human resources and machine technologies –uncertainty remains about their contribution to productivity growth and the prospects to intensify and unlock their potential to increase productivity. Economic policies restricting the market mechanism regulate the global labour market. Labour productivity shows moderate growth – the opportunities of the age of intelligent machines are entirely unrealised. Therefore, the price of labour productivity was a constant amongst OECD countries in 2015.

Acknowledgement: The paper is an output of the projects NFP313010BWN6 "The implementation framework and business model of the Internet of Things, Industry 4.0 and smart transport" and APVV-21-0188 "Development of concepts of strategic investment and development packages for regions of Slovakia with a causal link to the creation of methodology for quantifying their effectiveness and efficiency".

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https://doi.org/10.1016/j.jik.2023.100407
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GDP per hour worked in 2001 was $84.0 but had increased in 2021 to $107.86 (OECD, 2022).

The literature gap consists of the uncertainty of the role and value of human resources in the growth of labour efficiency in the age of intelligent machines. Actual research proposes only general theoretical assumptions on the reduced contribution of human resources to labour efficiency. Due to insufficient empirical research, i.e., weak evidence base, uncertainty emerges regarding the contribution of human resources and machine technologies to productivity growth.

In this paper, the research focuses on the future of global labour markets and workplaces. This paper aims to determine how automatisation reshapes the workplace. The main theoretical implications of the research lie in the vital role and significant value of human resources in the growth of labour efficiency. Future labour markets and workplaces are linked to labour conditions by redistributing functions between intelligent machines for complex and routine activity and human resources for creative and knowledge-intensive labour in decent workplaces. The research implications are connected with the prospects of improving human resources management. Reducing human resources' participation is related to a flexible, functional load on human resources and comfortable conditions for unlocking human potential and labour efficiency growth.

The research question of this paper is as follows: How do innovation and digitalisation reshape the workplace? Moreover, what are the consequences of this for labour efficiency? The research hypothesises that the age of intelligent machines has just begun; therefore, machine technologies are characterised by moderate automation, maintaining their dependence on human resources. In the current age of intelligent machines, productivity growth is the attraction of foreign human resources (a key source) through the global labour market. The pressure of government regulation must be eased and give way to competition and marketing.

To answer these research questions and verify the hypothesis, we specify the causal connections of the development of the global labour market and model output per worker on global robot distribution and the ease of hiring foreign labour, as well as on active labour market policies, pay, and productivity. The article aims to model the regularity of labour productivity growth in future workplaces depending on human resources and machine technologies and develop recommendations on productivity increases by managing competition and marketing in the global labour market.

The article's uniqueness lies in modelling global labour markets and gauging human resources and machine technologies' contribution to labour productivity growth in future workplaces. We achieve this by quantifying their potential to promote productivity growth and set clear economic policy guidelines by prioritising human resources and machine technology. Labour productivity is determined by human resource availability, not through automation. The study's novelty lies in contrasting alternative approaches to managing the development of the global labour market: government regulation (the first approach) and competition and marketing (the second approach). This offers detailed applied recommendations for economic policy.

This paper provides a new vision of reshaping workplaces under automatization (innovation and digitalization). In this new vision, machine technologies do not replace but supplement human resources in the age of intelligent machines. This is an essential difference compared to the existing literature, which makes this paper helpful in improving the effectiveness of intelligent machines and the organization of workplaces. The following idea was offered: human resources management remains essential under the conditions of the Fourth Industrial Revolution and requires transformation: an addition of support for human resources adaptation to the expanding automatization.

The article includes the following: 1) Modelling the global labour market and future workplaces through the measurement of the contribution of human resources and machine technologies on labour productivity; 2) Factor analysis of state regulation and de-regulation based on competition and marketing; 3) Policy implications for the development of the global labour market and workplaces.

## Literature review

### *Theory of the future of workplaces*

The article draws on the theory of the future of workplaces, which consists of the alternative character of future conditions and content of labour and organisation of workspace dependence on the course of the Fourth Industrial Revolution. This theory also implies a close interconnection between labour efficiency and automatisation. According to this theory, international labour and the latest trends in the global labour market in current economic conditions are found in Diab (2020); Han (2020); McGann, Murphy and Whelan (2020); Yadav (2020) and Zhang et al., (2020).

Productivity is justified as an assessment criterion for the development of the global labour market, as per Agbahey et al. (2020), Grodzicki and Skrzypek (M.J. 2020), Haeussler and Sauermann (C. 2020), Matthess and Kunkel (M. 2020), Rikap and Flacher (2020); Robinson (2020); Shahen et al. (2020) and Vecchiato (2020). The specifics of the functioning of labour markets are studied by Al-Ubaydli and List (2019), Bolíbar (2020), Mair (2018), Moro, Frank, Pentland, Rutherford, Cebrian and Rahwan (2021), Ricciardi et al. (2021) and Zhang et al. (2019). Finally, the fundamental issues of human resources management are explored by Angrist et al. (2021), Levy et al. (2021) and Soreq et al. (2021).

The literature review showed that the literature indicates a severe effect of innovation and digitalisation on workplaces. The future of global labour markets and workplaces depends on the distribution of roles between human resources and intelligent machines.

### *The global labour market in the age of intelligent machines*

A qualitative interpretation of the sources of development of the global labour market – human resources and machine technologies – as well as the aspects of the organisation and future workplaces, are given in the works of Bertello et al. (2021b), Bresciani et al. (2021a), Bresciani et al. (2021b), Coff et al. (2020), Haynes (2020), López-Cabarcos et al. (2020), Skare and Soriano (2021), Wan (2019), Yang and Gan (2020) and Zong et al. (2020). Selected competition and marketing issues in the global labour market are reflected in Felix (2020), Man (2020), and Shook et al. (2020).

The existing literature provides mixed evidence on the extant competition between human resources and machine technologies. Nevertheless, their contribution to productivity growth is not precisely determined (span 1). Moreover, the features of competition and marketing in the global labour market in intelligent machines have not yet been sufficiently studied. The importance of their relationship for labour productivity growth compared to government regulation has not been revealed (gap 2). The need to fill these gaps dictates a systematic modelling of future labour markets and workplaces, considering both sources of human resources and machine technologies in the context of government regulation and competition. Consequently, this research aims to fill these gaps.

The extant literature addresses the future of workplaces. Gianecchini et al. (2022) note that future labour is shaped by the pace of technological progress, accelerated by the impact of the Fourth Industrial Revolution. Similarly, Erickson and Norlander (2022) highlight that previous outsourcing and offshoring trends would be transformed into a future of remote work after the pandemic. Levasseur et al. (2022) also identified the strong influence of venture capital priorities on the future of labour.

Geyman (2022) argues that the future of labour in America lies in the decline of employer-sponsored insurance and the need to find a replacement. In turn, Iwashita (2021) proves that the consequences of COVID-19 can be associated with a decrease in the effectiveness of international personnel management, which determined the future of remote work in Japan. Jeffrey (2021) argues that automation will define the future of labour. On the other hand, Makridis and Han (2021), in examining decades of technological change, conclude that the future of labour is about employee empowerment and job satisfaction requirements. Donley (2021) substantiates the strong impact of the work environment on job satisfaction in a pre-pandemic database.

The analysis of the existing literature showed that it acknowledges and emphasises changes in the global labour market. These changes are directly connected with the use and management of human resources and thus require a new organisation of workplaces.

*Gap analysis*

The content analysis of the available literature showed that it focuses on disclosing the potential of using advanced technologies. At the same time, it failed to comprehensively explore the social consequences, from slightly different from the modern workplace model with an inevitable increase in the share of automated labour processes to a model of total automation and autonomous "smart" industries. The uncertainty (blurring) of the outlines of the model for future workplaces is another gap in the literature (gap 3).

Drawing on organising future workplaces, there are market features (e.g., the impossibility of fully automating workplaces in those processions where communications with customers are supposed) and technology limits (different possibilities for automating various business processes). This article proposes a new basis for choosing a future model of workplace organisation and considers the interests of entrepreneurs and employees of Industry 4.0. The proposed new model for organising future workplaces allows for shifting the focus from technology to workers and ensures a balanced and efficient sharing of human and technological resources – technological and social systems working in harmony – in Industry 4.0.

As a result of this literature review, it is possible to conclude that the theory of the future of workplaces connects the prospect of improving labour organisation with the dissemination of intelligent machines. The contribution of intelligent machines to improving workplace organisation remains unclear, and the consequences of automatisation could be contradictory.

The works by Baird et al. (2022), Davenport and Euchner (T. 2023), Fossa (2023) and Peng et al. (2023) indicate that the age of intelligent machines just started. Therefore, machine technologies are characterised by moderate automatisation, retaining their dependence on human resources.

The following hypothesis H is proposed: Labour efficiency is determined by human resources.

With the reliance on the materials of the official statistics, the proposed hypothesis acquires the following form: output per worker (GDP constant 2011 international $ in PPP) in 2019 positively depends on the ease of hiring foreign labour in 2020. To test this hypothesis, we use the method of regression analysis and find the regression dependence of the output per worker (GDP constant 2011 international $ in PPP) in 2019 on the ease of hiring foreign labour in 2020.

## Materials and methods

This article quantitatively measures and composes a competition model between human resources and intelligent machines with a systematic assessment of the factors they depend on and their influence on labour productivity. In the first stage, we employ a regression equation of the form:

$$y = a + b_1 * x_1 + b_2 * x_2$$

We regress the dependence of labour productivity in constant 2011 prices (y) according to the International Labour Organization (M. 2020) on the potential sources of its increase through the development of the global labour market, namely, the global distribution of robots ($x_1$) according to IMD (M. 2020) and the availability of foreign human resources as a manifestation of international labour migration ($x_2$) according to the World Economic Forum (M. 2020). We used 2011 as a baseline because, firstly, this is the form of the official statistics of the International Labour Organization (M. 2020). Secondly, 2011 belongs to the period of transition. Thus, the statistics on the digital competitiveness of IMD (M. 2020) have been collected since 2013. Therefore, 2011 is the beginning of the age of intelligent machines, the starting point at which the measuring of the change of labour efficiency in the current stage of intelligent machines should be applied. The second stage of the research is a regression equation of the form:

$$x_1 = c_1 + d_{11} * f_1 + d_{12} * f_2 \text{ and } x_2 = c_2 + d_{21} * f_1 + d_{22} * f_2.$$

This determines the regression dependence of the global distribution of robots ($x_1$) and the availability of foreign human resources ($x_2$) on active labour market policies (factor of state regulation of the labour market, $f_1$) and the connection of wages with productivity (factor of competition and marketing in the labour market, $f_2$). The values of both factors ($f_1$ and $f_2$) are taken from the World Economic Forum (M. 2020).

Since the global distribution of robots is measured by the IMD (M. 2020) in ranking positions (the higher the ranking, the better) and all other indicators in points (the larger, the better), the positive effect of the $x_1$ factor on y will be evidenced by a negative regression coefficient ($b_1<0$). In all other cases, the values of regression coefficients greater than 0 indicate a positive effect. It follows that H is true if all conditions are met:

$$\begin{cases} b_2 > b_1; \\ d_{11} > d_{12}; \end{cases}$$

Our representative selection includes three countries from each category according to the level of economic freedom per the rating and classification of The Heritage Foundation (2022), except for "Repressed" and "Non-Ranked", for which the values of the indicators are not available. We give the economic freedom index values in a selection of countries in 2022 in Fig. 1.

The representative selection of countries is as follows: 1) Different levels of economic freedom, from high, e.g., in Ireland (82.0 points) – to moderate, e.g., in China (48.0 points); 2) Different levels of income according to the classification by the World Bank (T. 2023) for 2022 –2023 (e.g., high income – Australia, upper middle – Russia, lower middle - India); 3) Different levels of socio-economic development (the sample contains developed countries of the OECD – Australia, Ireland, and developing countries, in particular, BRICS countries.

The prospects of global labour markets amidst intelligent machines are determined by optimising factors (contrasting human resources and machine technologies) and the sources of their initiation (government regulation, competition, and marketing). Statistics for 2020 are illustrated in Table 1.

The multicollinearity test is provided in Table 2.

As shown in Table 2, neither of the coefficients of cross-correlation of the considered variables exceeded the value of 0.90.

## Results

Modelling the global labour market and workplaces of the future in the age of intelligent machines by measuring the contribution of human resources and machine technologies to labour productivity

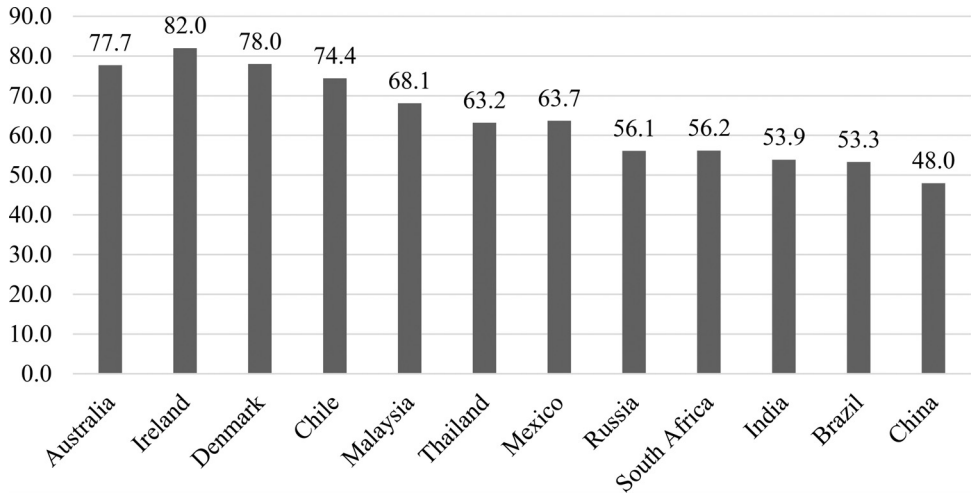



**Fig. 1.** Index of economic freedom in the selected countries in 2022, points 1–100. Source: Authors' elaboration based on The Heritage Foundation (2022)

The statistical approach relies on statistics from The Heritage Foundation, the IMD, and the World Economic Forum, amongst others. We cover a representative sample in which countries with varying levels of economic freedom are included. We modelled the global labour market by measuring human resources and machine technologies to labour productivity, as shown in Fig. 2.

Productivity increases along with the expanded availability of human resources but decreases as the use of robots spreads across the globe ($b_2>b_1$) (see Fig. 2). The global spread of robots does not contribute to developing the worldwide labour market and will not be further considered in this article. The regression model refines the global labour market by measuring human resources' contribution (see Fig. 3).

According to Fig. 3, a 1-point increase in human resources availability will result in a $1024.60 increase in labour productivity (at constant prices in 2011). Organizing future workplaces implies limited automatisation to effectively preserve the key role of human resources.

Factor analysis of state regulation and de-regulation on competition and marketing in the development of the global labour market in the age of intelligent machines

A factor analysis of state regulation and de-regulation in the era of smart machines is displayed in Fig. 4.

According to Fig. 4, the availability of human resources decreases as government regulation increases and competition and marketing-based de-regulation increase ($d_{21}>d_{22}$). We thus present a refined regression model of the global labour market through the dependence of the availability of human resources on competition and marketing in Fig. 5.

According to Fig. 5, a 1-point increase in competition and marketing will result in a 0.3765-point increase in the availability of human resources.

Policy implications for developing the worldwide labour market and future workplaces in the age of intelligent machines based on competition and marketing

We turn to solve the optimisation problem ($y\rightarrow max$). There are, however, limitations in such optimisation. Specifically, the availability of foreign human resources should not exceed 100 points ($x_2\leq100$), and the association of remuneration with productivity should not exceed 100 points ($f_2\leq100$). The simple substitution method yields the following result (Fig. 6).

According to Fig. 6, the prospects for optimising the global labour market suggest a necessary increase in labour productivity (in constant prices in 2011) by 24.78% (from 66.04 in 2020 to 82.40). The following policy implications apply:

**Table 1**
Statistics of the global labour market in the age of intelligent machines in 2020.

| – | – | State labour market regulation factor | Competition and marketing factors | Provisional results | | Result |
|---|---|---|---|---|---|---|
| Economic freedom | Country | Active labour market policies, points 1–100 | Pay and productivity, points 1–100 | World robot distribution, position 1–63 | Ease of hiring foreign labour, points 1–100 | Output per worker (GDP constant 2011 international $ in PPP), 2019 |
| | | $f_1$ | $f_2$ | $x_1$ | $x_2$ | y |
| Free | Australia | 64.5 | 58.1 | 29 | 32.8 | 94,644 |
| | Ireland | 61.4 | 62.4 | 43 | 62.8 | 155,654 |
| Mostly free | Denmark | 70.6 | 65.1 | 30 | 45.0 | 97,696 |
| | Chile | 34.4 | 56.0 | 48 | 58.9 | 49,464 |
| | Malaysia | 64.7 | 70.3 | 22 | 63.0 | 61,291 |
| Moderately free | Mexico | 29.3 | 46.4 | 10 | 56.7 | 41,554 |
| | Russia | 44.0 | 58.9 | 32 | 48.1 | 52,971 |
| | Thailand | 46.3 | 60.1 | 11 | 52.5 | 31,204 |
| | China* | 58.2 | 60.5 | 1 | 58.0 | 13,800 |
| Mostly unfree | South Africa | 24.9 | 46.0 | 34 | 40.6 | 42,210 |
| | India | 41.8 | 51.3 | 12 | 44.4 | 21,181 |
| | Brazil | 27.4 | 40.4 | 17 | 43.9 | 32,232 |

*As classified by the authors of this paper, China is assigned to the "moderately free" category.
Source: Authors' elaboration.

**Table 2**
Multicollinearity test.

| | f1 | f2 | x1 | x2 | y |
|---|---|---|---|---|---|
| f1 | 1 | 0.8729 | 0.0464 | 0.1248 | 0.5618 |
| f2 | – | 1 | 0.1635 | 0.4107 | 0.4357 |
| x1 | – | – | 1 | 0.0031 | 0.6455 |
| x2 | – | – | – | 1 | 0.0899 |
| y | – | – | – | – | 1 |

Source: Authors' elaboration.

- Increase the availability of foreign human resources by 31.72% (from 50.35 points in 2020 to 66.32 points).
- This increases wages with productivity by 73.69% (from 57.58 points in 2020 to 100 points).

Since we researched a representative selection of countries, our policy implications might be universal for countries with any level of economic freedom. Thus, the organisation of workplaces of the future, which stimulates the growth of labour efficiency, can be outlined as follows:

- Use of human resources as the subject of labour during the creation of international workgroups and the use of international project teams.
- Active use of intelligent machines as a tool of labour.
- Use of intelligent machines (i.e., machine vision) to assess the individual results of each employee's labour for stimulation of "healthy" competition of human resources.
- Intensive use of the mechanisms of material stimulation of employees' labour (within marketing human resources) to support competition of human resources.

## Discussion

The article contributes to developing the theory of the future of workplaces, specifying the character of the influence of machine technology on labour efficiency in labour markets and workplaces. In

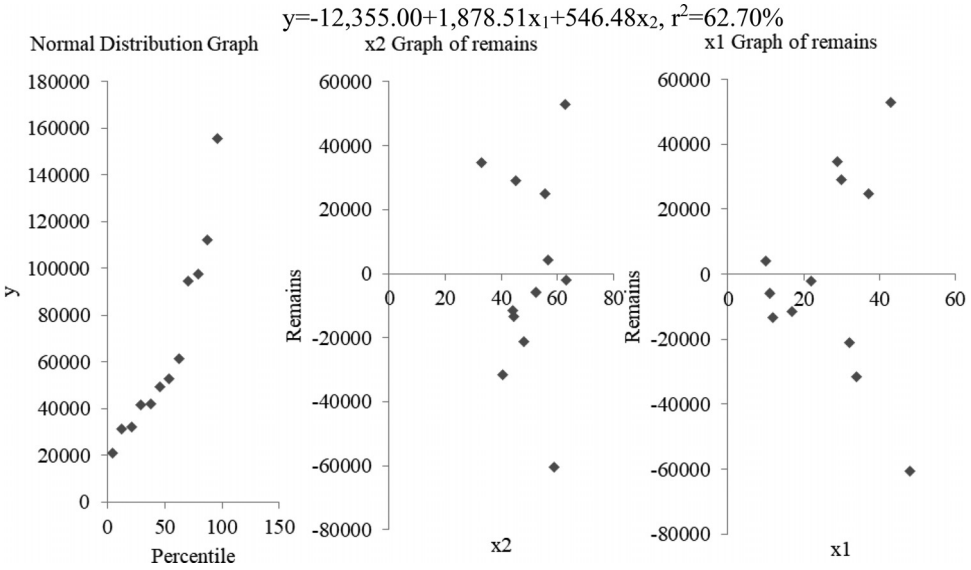



**Fig. 2.** Regression statistics of the dependence of labour productivity on human resources and machine technologies. Source: Authors' elaboration

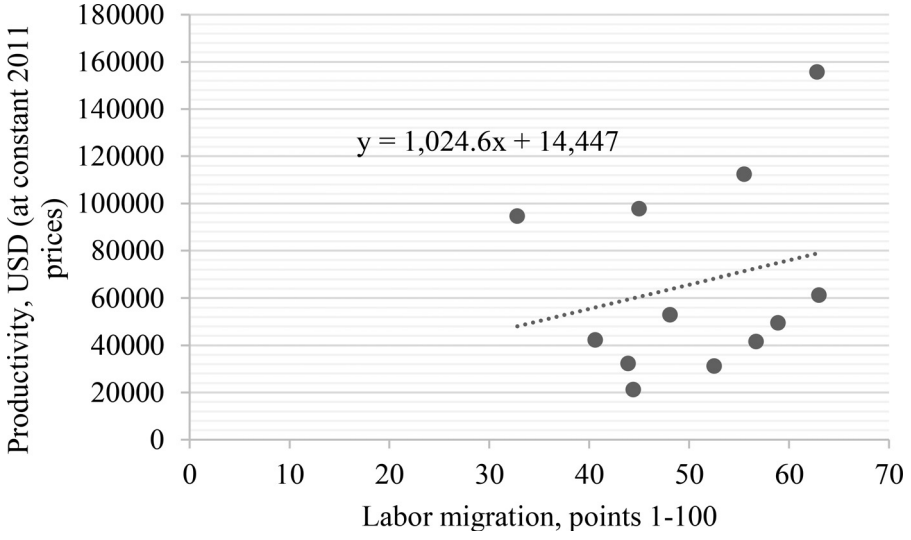



**Fig. 3.** Regression curve of labour productivity versus human resources in 2020. Source: Authors' elaboration

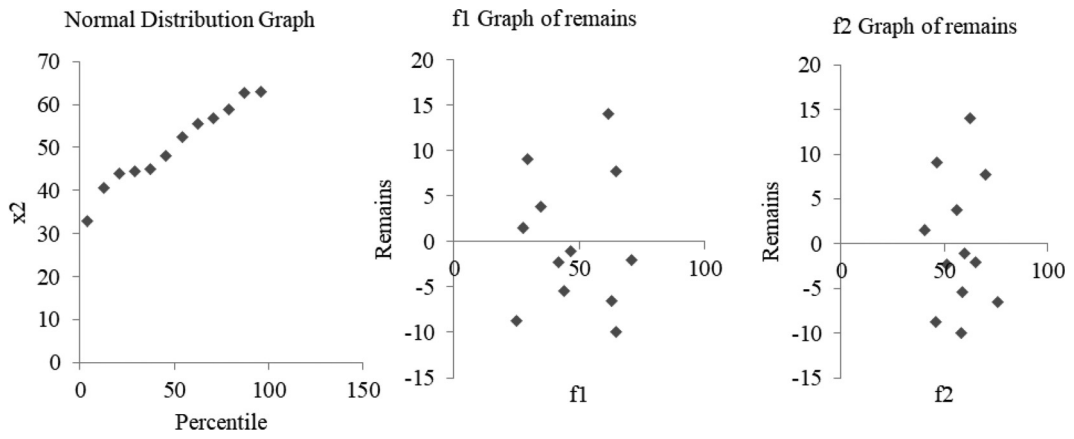



**Fig. 4.** Regression statistics of the dependence of human resources availability on government regulation and deregulation based on competition and marketing. Source: Authors' elaboration

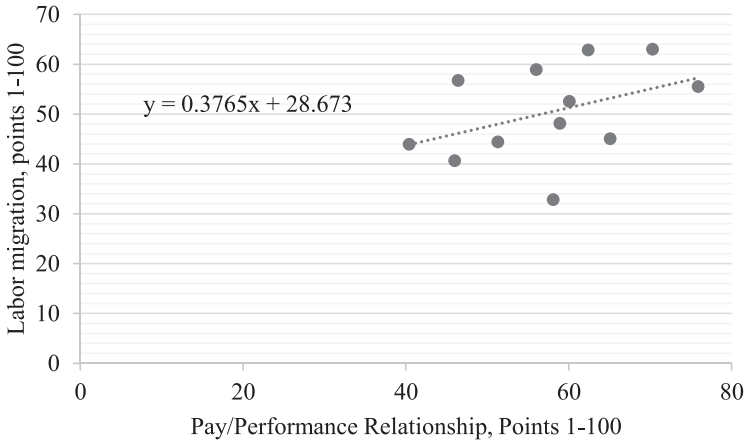
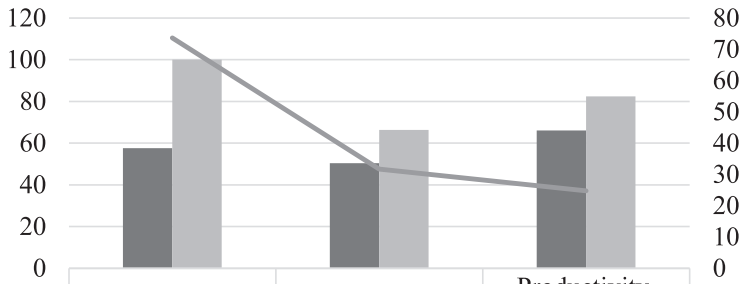



**Fig. 5.** Regression curve of human resource availability versus competition and marketing. Source: Authors' elaboration

contrast to the existing literature, e.g., Donley (2021), Erickson and Norlander (2022), Geyman (2022), Gianecchini et al. (2022), Iwashita (2021), Jeffrey (2021), Levasseur et al. (2022), and Makridis and Han (2021), this article's findings revealed that the future of workplaces is determined not by market features and technology limits but by the benefits of different workplace models. The article's new basis for choosing various models for organising future workplaces has subsequent implications for productivity. The advantages of the new base compared to the old base are illustrated in Table 3.



| | Pay and Productivity, points 1-100 | Ease of hiring foreign labour, points 1-100 | Productivity, USD thousand (at constant 2011 prices) |
|---|---|---|---|
| Baseline values (2020) | 57.58 | 50.35 | 66.04 |
| Prospects for the progress of values | 100 | 66.32 | 82.4 |
| Recommended progress, % | 73.69 | 31.72 | 24.78 |

**Fig. 6.** Prospects for optimising the development of the global labour market in the age of intelligent machines based on competition and marketing. Source: Authors' elaboration

**Table 3**
Advantages of the new basis for choosing a model for organising future workplaces.

| Comparison criterion | Existing literature | New results obtained from the article | Benefits of new results |
|---|---|---|---|
| The basis for choosing a model for organising future workplaces | Market specifics and technology limits | Productivity implications | Focus not on inputs (causes, opportunities) but on outputs (consequences) |
| Interests are considered when choosing a model for organising future workplaces. | Business interests only | Both the interests of business and the interests of employees | Benefits for all stakeholders, as well as systemic social, economic, and technological crisis management in the organisation of future workplaces |
| Social implications of the workplace of the future | Technophobes, who fear or dislike technological development | Technophiles, who stand in defence of new technologies | Technological and social systems work in harmony and will support future technological progress |
| Consequences of the transition to new models of organisation of workplaces of the future for human potential | Limitation of opportunities for the development and realisation of human potential | Expanding opportunities for the development and realisation of human potential | Achieving a synergistic effect in the form of the complete realisation of human potential with systemic social and technological progress |

Source: Authors' elaboration.

This paper clarified the causal relationships behind increases in labour productivity by exploring the consequence of workplace optimisation. Thus, the results of this research can contribute to ongoing discussions within the studied topics of the organisation of workplaces of the future in Agbahey et al. (2020), Grodzicki and Skrzypek (M. J. 2020), Haeussler and Sauermann (C. 2020), Matthess and Kunkel (M. 2020), Rikap and Flacher (2020); Robinson (2020); Shahen et al. (2020) and Vecchiato (2020).

As demonstrated in Table 3, one of the advantages of choosing the proposed basis is the focus not on inputs (causes, opportunities) but on outputs (consequences). This results in a transition from "technology for technology's sake" to seeking to benefit from new technologies and considering that future workplaces are not impersonal business process models but future working conditions for each individual. In this regard, the theory of the future of workplaces becomes multidisciplinary and covers both management science and social sciences. This opens up a wide field for further research and facilitates a deeper analysis of the theories of the future of workplaces.

Another advantage of the new basis for making managerial decisions on reshaping the workplace under innovation and digitalisation is that it systematically considers the consequences for both businesses and employees. The new foundation stimulates the desire to benefit all stakeholders. It provides systemic social, economic, and technological crisis management in future jobs.

The social consequences of the organisation of future workplaces under the old basis are associated with transforming modern workers into technophobes who fear or dislike technological development. The new basis facilitates the creation of a society of technophiles who defend new technologies. This new basis will result in technological and social systems that work in harmony. Furthermore, social support will be provided for future technological progress and avoid massive resistance to change and undermining of technological innovations on the part of workers.

The consequences of the transition to new models of organising workplaces of the future for human potential under the old basis are associated with limited opportunities for developing and realizing human potential. Under the new ground, these consequences will be increased opportunities for growth and human potential. A synergistic effect will be achieved in realising human potential with systemic social and technological progress. The focus will be shifted from technology to workers to ensure the balanced and efficient sharing of human and technical resources.

The COVID-19 outbreak further amplifies the complexity of transitioning to future workplace models (Bertello et al., 2021a; Kraus et al., 2020). Indeed, social tensions seem to have reached their limit. Many workers have lost their jobs or learned new skills and filled digital competency gaps when moving to remote work (Wendt et al., 2021). Consequently, it is imperative to mitigate the social consequences of reshaping workplaces under the influence of innovation and digitalisation. Moreover, the new basis for decision-making will make it possible to use disruptive technological advancements to relieve social tension in the post-pandemic period.

The new basis will also support HR-related practices to select, motivate, develop, and retain employees. Firms' efforts to align strategies with their current employees' skills will provide them with reskilling and upskilling programmes and support their ability to manage people. The future of the workplace depends on the (re)actions of individuals and firms, as well as their adaptability to new technological scenarios; the new basis will thus make this possible. The policy implications of organisational behaviour and human resource management research in response to disruptive technological development are related to stimulating Industry 4.0 enterprises to increase labour productivity.

## Conclusions

Labour productivity remains on human resource availability (not through automation). The global spread of machine technologies does not contribute to productivity growth, presumably under the long-term influence of the third law of philosophy. While there will be a transition from quantitative to qualitative, this has not yet happened, and tactical performance management. The lack of digital competencies by robot automation personnel can explain this; robots can be misused, slowing production. In the medium term, we associate this result with the prospects for developing the global labour market with increased human resource availability, as determined by de-regulation. The mass training of digital personnel will overcome this problem in the long term, at least with a shortage of digital personnel, increasing the efficiency of managing the global labour market. Reliance on competition and marketing will also maximise labour productivity in the medium term (until 2023).

The current results and conclusions of the article include the rationale that human resources make a significant and invaluable contribution to the increase in labour productivity. Despite the ongoing Fourth Industrial Revolution, the impact of robots' expansion on productivity in the modern age of intelligent machines is small and controversial. Therefore, it is advisable to focus on human resources development at the current stage of the Fourth Industrial Revolution. Second, contrary to the prevailing idea of state regulation of labour markets, the effective management of human resources in private entrepreneurship more extensively promotes the development of human resources.

Applied recommendations and benchmarks explain the importance of the results for policymaking, particularly for crucial

statistical indicators that improve labour productivity management. The guidance can apply in diverse contexts. Standards for key statistical indicators that enhance labour productivity management explain the importance of the results for policymaking. The proposed concept effectively demonstrates that innovation and digitalisation reshape the workplace. Although developing machine technologies stretched beyond the focus of this research, they could likely become the future growth vector of global labour markets, providing mass availability and high-quality digital personnel.

This paper's implications include clarifying the essence of reshaping workplaces through innovation and digitalisation. The main theoretical conclusion from the research is that machine technologies do not replace to complement human resources. Due to this, the paper contributes to and supports the ongoing productive discussions – in particular, in the works of Baird et al. (2022), Davenport and Euchner (T. 2023), Fossa (2023) and Peng et al. (2023) – with new evidence on the issues of the future organisation of workplaces, strengthening of the evidence base of human resources remaining and preserving its role in entrepreneurial processes. Therefore, the management of human resources is still topical. However, it undergoes profound changes supplemented by adapting human resources to the expanding automatisation.

The article's implications include the prospects for labour division between human resources and machine technologies in future workplaces. Human resource management implications of this research are creating equally favourable conditions for further automatisation and unlocking human potential. The appearance of a new phenomenon of intelligent machines and hybrid workplaces shows adaptability to comfortable human labour and the use of the leading automation tools – intelligent machines.

A limitation of the findings is the study's focus on the organisation of workplaces and results for productivity, while the psychology of labour requires further in-depth research. The phenomenon of hybrid workplaces requires further elaboration. Future studies should dwell on the experience of workplaces and a high level of automatisation. Identifying alternative forms of hybrid workplaces in various labour conditions would be expedient.

## References


Agbahey, J., Siddig, K., & Grethe, H. (2020). Implications of labour supply specifications in CGE models: A demonstration for employment of Palestinian labor in Israel and its impact on the West Bank economy. *Economic Analysis and Policy, 68*, 265–284.

Al-Omoush, K. S., Simón-Moya, V., & Sendra-García, J. (2020). The impact of social capital and collaborative knowledge creation on e-business proactiveness and organizational agility in responding to the COVID-19 crisis. *Journal of Innovation and Knowledge, 5*(4), 279–288. doi:10.1016/j.jik.2020.10.002.

Al-Ubaydli, O., & List, J. A. (2019). How natural field experiments have enhanced our understanding of unemployment. *Nature Human Behaviour, 3*, 33–39. doi:10.1038/s41562-018-0496-z.

Angrist, N., Djankov, S., Goldberg, P. K., & Patrinos, H. A. (2021). Measuring human capital using global learning data. *Nature, 592*, 403–408. doi:10.1038/s41586-021-03323-7.

Baird, I., Fendley, M. E., & Warren, R. (2022). The human-automation behavioral interaction task (HABIT) analysis framework. *Human Factors and Ergonomics In Manufacturing, 32*(6), 452–461. doi:10.1002/hfm.20963.

Ballestar, M. T., Camiña, E., Díaz-Chao, Á., & Torrent-Sellens, J. (2021). Productivity and employment effects of digital complementarities. *Journal of Innovation and Knowledge, 6*(3), 177–190. doi:10.1016/j.jik.2020.10.006.

Bertello, A., Bogers, M. L., & De Bernardi, P. (2021). Open innovation in the face of the COVID-19 grand challenge: Insights from the Pan-European hackathon 'EUvsVirus. *R&D Management*. doi:10.1111/radm.12456.

Bertello, A., Ferraris, A., Bresciani, S., & De Bernardi, P. (2021). Big data analytics (BDA) and degree of internationalization: The interplay between the governance of BDA infrastructure and BDA capabilities. *Journal of Management and Governance, 25*, 1035–1055.

Bolíbar, M. (2020). Social capital, human capital and ethnic occupational niches: An analysis of ethnic and gender inequalities in the Spanish labour market. *Palgrave Communications, 6*(22). doi:10.1057/s41599-020-0397-4.

Boza, I., & Ilyés, V. (2020). Decomposition of co-worker wage gains. *IZA Journal of Labor Economics, 9*(1). doi:10.2478/izajole-2020-0008.

Bresciani, S., Ferraris, A., Romano, M., & Santoro, G. (2021). *Human resource management and digitalisation. in digital transformation management for agile organizations: A compass to sail the digital world*. Bingley, UK: Emerald Publishing Limited.

Bresciani, S., Huarng, K. H., Malhotra, A., & Ferraris, A. (2021). Digital transformation as a springboard for product, process and business model innovation. *Journal of Business Research, 128*, 204–210.

Camargo, J., Lima, L., Riva, F., & Souza, A. P. (2021). Technical education, non-cognitive skills and labour market outcomes: Experimental evidence from Brazil. *IZA Journal of Labor Economics, 10*(1). doi:10.2478/izajole-2021-0002.

Coff, R., El-Zayaty, A., Ganco, M., Mawdsley, J. K., Tzabbar, D., & Cirillo, B. (2020). Firm-specific human capital at the crossroads: A conversation on current issues and future directions. Eds., *Employee inter- and intra-firm mobility (advances in strategic management, vol. 41)* Eds.. (pp. 55–73). Bingley, UK: Emerald Publishing Limited. doi:10.1108/S0742-332220200000041003.

Corella, L. F. M. (2020). Minimum wages in monopsonistic labour markets. *IZA Journal of Labor Economics, 9*(1). doi:10.2478/izajole-2020-0007.

Davenport, T., & Euchner, J. (2023). The rise of human-machine collaboration: An interview with Tom Davenport. *Research Technology Management, 66*(1), 11–15. doi:10.1080/08956308.2023.2142435.

Diab, A. A. A. (2020). Interplay between labour dynamics, accounting and accountability practices during the rise of a political logic: An Egyptian case study. *Qualitative Research in Accounting & Management, 17*(4), 675–702. doi:10.1108/QRAM-12-2019-0134.

Donley, J. (2021). The impact of work environment on job satisfaction: Pre-COVID research to inform the future. *Nurse Leader, 19*(6), 585–589. doi:10.1016/j.mnl.2021.08.009.

Erickson, C. L., & Norlander, P. (2022). How the past of outsourcing and offshoring is the future of post-pandemic remote work: A typology, a model and a review. *Industrial Relations Journal, 53*(1), 71–89. doi:10.1111/irj.12355.

Felix, C. V., Sengupta, E., Blessinger, P., & Makhanya, M. S. (2020). The role of the teacher and AI in education. *International perspectives on the role of technology in humanizing higher education (innovations in higher education teaching and learning)* (pp. 33–48). Bingley, UK: Emerald Publishing Limited. doi:10.1108/S2055-364120200000033003 Vol. 33.

Fossa, F. (2023). The high road. Driving automation, human values, and artificial agency. *Studies in Applied Philosophy, Epistemology and Rational Ethics, 65*, 139–148. doi:10.1007/978-3-031-22982-4_7.

Geyman, J. (2022). The future of work in America: Demise of employer-sponsored insurance and what should replace it. *International Journal of Health Services, 52*(1), 168–173. doi:10.1177/00207314211044002.

Gianecchini, M., Dotto, S., & Gubitta, P. (2022). Shaping the future of work. *Lecture Notes in Information Systems and Organisation, 49*, 67–83. doi:10.1007/978-3-030-83321-3_5.

Grodzicki, M. J., & Skrzypek, J. (2020). Cost-competitiveness and structural change in value chains – vertically-integrated analysis of the European automotive sector. *Structural Change and Economic Dynamics, 55*, 276–287.

Haeussler, C., & Sauermann, H. (2020). Division of labour in collaborative knowledge production: The role of team size and interdisciplinarity. *Research Policy, 49*,(6) 103987.

Han, J. (2020). How to promote rural revitalization via introducing skilled labour, deepening land reform and facilitating investment? *China Agricultural Economic Review, 12*(4), 577–582. doi:10.1108/CAER-02-2020-0020.

Haynes, N. C. (2020). Robots, artificial intelligence and service automation in travel, tourism and hospitality. *Journal of Tourism Futures, 6*(2), 191–192. doi:10.1108/JTF-06-2020-149.

Hober, B., Schaarschmidt, M., & von Korflesch, H. (2021). Internal idea contests: Work environment perceptions and the moderating role of power distance. *Journal of Innovation and Knowledge, 6*(1), 1–10. doi:10.1016/j.jik.2019.11.003.

IMD. 2020. World digital competitiveness ranking 2020. https://www.imd.org/research-knowledge/articles/the-imd-world-digital-competitiveness-ranking/. Accessed December 20, 2020.

International Labor Organization. 2020. Statistics on labour productivity. https://ilostat.ilo.org/topics/labour-productivity/. Accessed December 20, 2020.

Iwashita, H. (2021). The future of remote work in Japan: Covid-19's implications for international human resource management. *Entrepreneurial Business and Economics Review, 9*(4), 7–18. doi:10.15678/EBER.2021.090401.

Jeffrey, K. (2021). Automation and the future of work: How rhetoric shapes the response in policy preferences. *Journal of Economic Behavior and Organization, 192*, 417–433. doi:10.1016/j.jebo.2021.10.019.

Kraus, S., Clauss, T., Breier, M., Gast, J., Zardini, A., & Tiberius, V. (2020). The economics of COVID-19: Initial empirical evidence on how family firms in five European countries cope with the corona crisis. *International Journal of Entrepreneurial Behavior & Research, 26*(5), 1067–1092.

Levasseur, L., Johan, S., & Eckhardt, J. (2022). Mixed methods in venture capital research: An illustrative study and directions for future work. *British Journal of Management, 33*(1), 26–45. doi:10.1111/1467-8551.12514.

Levy, D. E., Pachucki, M. C., O'Malley, A. J., Pomeala, B., Yaqubi, A., & Thorndike, A. N (2021). Social connections and the healthfulness of food choices in an employee population. *Nature Human Behaviour, 5*(10), 1349–1357. doi:10.1038/s41562-021-01103-x.

López-Cabarcos, M.Á., Ribeiro-Soriano, D., & Piñeiro-Chousa, J. (2020). All that glitters is not gold. The rise of gaming in the COVID-19 pandemic. *Journal of Innovation & Knowledge, 5*(4), 289–296.

Mair, J. (2018). Scaling innovative ideas to create inclusive labour markets. *Nature Human Behaviour, 2*(884). doi:10.1038/s41562-018-0352-1.

Makridis, C. A., & Han, J. H. (2021). Future of work and employee empowerment and satisfaction: Evidence from a decade of technological change. *Technological Forecasting and Social Change, 173*, 121162. doi:10.1016/j.techfore.2021.121162.

Man, M. M. K., Turkmenoglu, M. A., & Cicek, B. (2020). Human resource development requirements in industrial revolution 4.0. *Contemporary global issues in human resource management* (pp. 129–139). Bingley, UK: Emerald Publishing Limited. doi:10.1108/978-1-80043-392-220201011.

Matthess, M., & Kunkel, S. (2020). Structural change and digitalization in developing countries: Conceptually linking the two transformations. *Technology in Society, 63*, 101428.

McGann, M., Murphy, M. P., & Whelan, N. (2020). Workfare redux? Pandemic unemployment, labour activation and the lessons of post-crisis welfare reform in Ireland. *International Journal of Sociology and Social Policy, 40*(9/10), 963–978. doi:10.1108/IJSSP-07-2020-0343.

(...) Moffett, M. W., Garnier, S., Eisenhardt, K. M., Bach, L. A., & Offenberg, J. (2021). Ant colonies: Building complex organizations with minuscule brains and no leaders. *Journal of Organization Design, 10*(1), 55–74. doi:10.1007/s41469-021-00093-4.

Moro, E., Frank, M. R., Pentland, A., Rutherford, A., Cebrian, M., & Rahwan, I. (2021). Universal resilience patterns in labour markets. *Nature communications, 12*, 1972. doi:10.1038/s41467-021-22086-3.

Organisation for Economic Co-operation and Development (OECD). 2022. GDP per hour worked, total, 2015=100, 2001–2021. https://data.oecd.org/lprdty/gdp-per-hour-worked.htm Accessed February 3, 2022.

Peng, C., Zhen, X., & Huang, Y. (2023). Human-automation interaction centered approach based on FRAM for systemic safety analysis of dynamic positioning operations for offshore tandem offloading. *Ocean Engineering, 267*, 113249. doi:10.1016/j.oceaneng.2022.113249.

Puranam, P. (2021). Human-AI collaborative decision-making as an organization design problem. *Journal of Organization Design, 10*, 75–80. doi:10.1007/s41469-021-00095-2.

Ricciardi, V., Mehrabi, Z., Wittman, H., James, D., & Ramankutty, N. (2021). Higher yields and more biodiversity on smaller farms. *Nature Sustainability, 4*, 651–657. doi:10.1038/s41893-021-00699-2.

Rikap, C., & Flacher, D. (2020). Who collects intellectual rents from knowledge and innovation hubs? questioning the sustainability of the Singapore model. *Structural Change and Economic Dynamics, 55*, 59–73.

Robinson, S. C. (2020). Trust, transparency, and openness: How the inclusion of cultural values shapes Nordic national public policy strategies for artificial intelligence (AI). *Technology in Society, 63*, 101421.

Schäfer, B., Koloch, L., Storai, D., Gunkel, M., & Kraus, S. (2023). Alternative workplace arrangements: Tearing down the walls of a conceptual labyrinth. *Journal of Innovation & Knowledge, 8*,(2) 100352. doi:10.1016/j.jik.2023.100352.

Schumacher, C. (2021). Organizational structure and CEO dominance. *Journal of Organization Design, 10*(1), 19–34. doi:10.1186/s41469-021-00091-6.

Shahen, M. E., Kotani, K., Kakinaka, M., & Managi, S. (2020). Wage and labour mobility between public, formal private and informal private sectors in a developing country. *Economic Analysis and Policy, 68*, 101–113.

Shook, J. R., Solymosi, T., Giordano, J., & Masakowski, Y. R. (2020). Ethical constraints and contexts of artificial intelligent systems in national security, intelligence, and defense/military operations. *Artificial intelligence and global security*, 137–152. doi:10.1108/978-1-78973-811-720201008.

Skare, M., & Soriano, D. R. (2021). How globalization is changing digital technology adoption: An international perspective. *Journal of Innovation & Knowledge, 6*(4), 222–233.

Soreq, E., Violante, I. R., Daws, R. E., & Hampshire, A. (2021). Neuroimaging evidence for a network sampling theory of individual differences in human intelligence test performance. *Nature Communications, 12*, 2072. doi:10.1038/s41467-021-22199-9.

The Heritage Foundation. 2022. 2020 Index of Economic Freedom 2022: Country Rankings. https://www.heritage.org/index/ranking. Accessed November 2, 2022.

Vecchiato, R. (2020). Analogical reasoning, cognition, and the response to technological change: Lessons from mobile communication. *Research Policy, 49*,(5) 103958.

Wan, H. L. (2019). Global human resources: A key to mission accomplishment. *Journal of Global Mobility, 7*(1), 5–26. doi:10.1108/JGM-01-2018-0007.

Wendt, C., Adam, M., Benlian, A., & Kraus, S. (2021). Let's connect to keep the distance: How SMEs leverage information and communication technologies to address the COVID-19 crisis. *Information Systems Frontiers, 24*, 1061–1079.

World Bank. 2023. New world bank country classifications by income level: 2022–2023. https://blogs.worldbank.org/opendata/new-world-bank-country-classifications-income-level-2022-2023. Accessed January 31, 2023.

World Economic Forum. 2020. The global competitiveness report 2019. https://www.weforum.org/reports/how-to-end-a-decade-of-lost-productivity-growth. Accessed December 20, 2020.

Yadav, C. L. (2020). An introduction of cultural rights of labour in Maritime employment in India. *International Journal of Law and Management, 62*(2), 139–145. doi:10.1108/IJLMA-03-2019-0064.

Yang, L., & Gan, C. (2020). Cooperative goals and dynamic capability: The mediating role of strategic flexibility and the moderating role of human resource flexibility. *Journal of Business & Industrial Marketing, 36*(5), 782–795. doi:10.1108/JBIM-11-2019-0495.

Zhang, D., Zhang, Q., Qi, S., Huang, J., Karplus, V. J., & Zhang, X. (2019). Integrity of firms' emissions reporting in China's early carbon markets. *Nature Climate Change, 9*, 164–169. doi:10.1038/s41558-018-0394-4.

Zhang, P., Zhao, L., Vata, O., & Rajagopal, S. (2020). Restructuring seafarers' welfare under the Maritime labour convention: An empirical case study of Greece. *Maritime Business Review, 5*(4), 373–389. doi:10.1108/MABR-02-2020-0009.

Zong, C., Ji, Z., Yu, J., & Yu, H. (2020). An angle-changeable tracked robot with human-robot interaction in unstructured environments. *Assembly Automation, 40*(4), 565–575. doi:10.1108/AA-11-2018-0231.